

Basic Principles of Medicine 2
Module: Digestion and Metabolism
DLA

**Histological Overview of the Salivary
Glands**

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Objectives

Differentiate between the microscopic structures of the three major salivary glands.	SOM.MK.I.BPM2.3.DM.1.HCB.1188
Identify the microscopic structure of the ducts of the submandibular gland.	SOM.MK.I.BPM2.3.DM.1.HCB.1189
Identify the microscopic structure of the ducts of the parotid gland.	SOM.MK.I.BPM2.3.DM.1.HCB.1190
Give the function(s) of saliva.	SOM.MK.I.BPM2.3.DM.1.HCB.1191
Describe the composition of saliva.	SOM.MK.I.BPM2.3.DM.1.HCB.1192

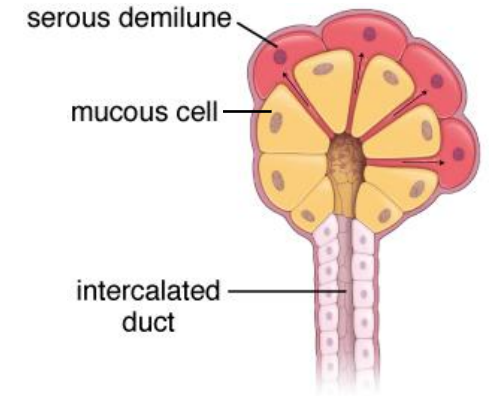
Review the last 3 slides from the Epithelium and Glands lecture from your FTM module in BPM 1 before reviewing this DLA.

Saliva

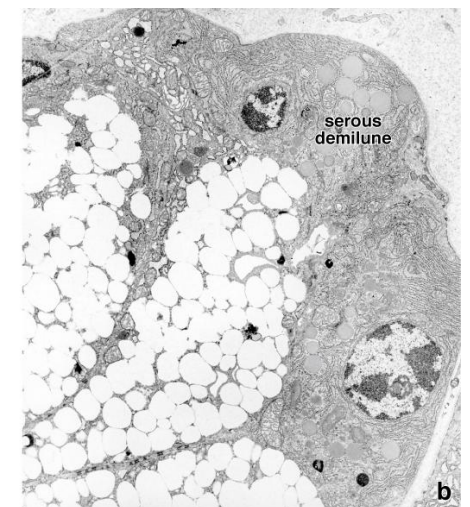
- Average daily production of saliva is 1.0 to 1.2 liters
- **Function**
 - Moistens and lubricates food: mucus and electrolytes
 - Protects the oral cavity by lubrication, control of bacterial flora (by the presence of lysozyme, peroxidase, histatin, cystatin, lactoferrin, and IgA), and its cleansing action.
 - Assists in taste sensation
 - Initiates digestion of carbohydrates via salivary amylase
 - Weak lipase to digest some fat
- **Composition**
 - Initial acinar secretion resembles tissue fluid
 - Modified by the ductal system
 - **Intercalated ducts** – adds bicarbonate and removes chlorine ions
 - **Straited ducts** – removes sodium and pumps potassium and bicarbonate into the saliva
 - Final secretion is hypotonic
 - Microbial flora and immune cells
 - FYI, also contains: Opiorphin (pain killer) haptocorrin (Vit B12 binding) and Kallikrein (forms bradykinin)

Salivary Acini (secretory units)

- Blind sac containing secretory cells
- Major salivary gland acini surrounded by a capsule of connective tissue, form septa divides the glands into lobes and lobules
- Three basic types:
 - **Serous**: contains serous/protein secreting cells.
 - **Mucous**: contains mucous cells
 - **Mixed**: contains both serous and mucous cells. The serous cells are arranged in a cap around the mucous cells called a serous demilune
- Myoepithelial cells:
 - Contractile cells that are located on the basal aspect of the acini and the proximal part of the ducts
 - Contraction helps move the secretory product out of the acini and into the ducts

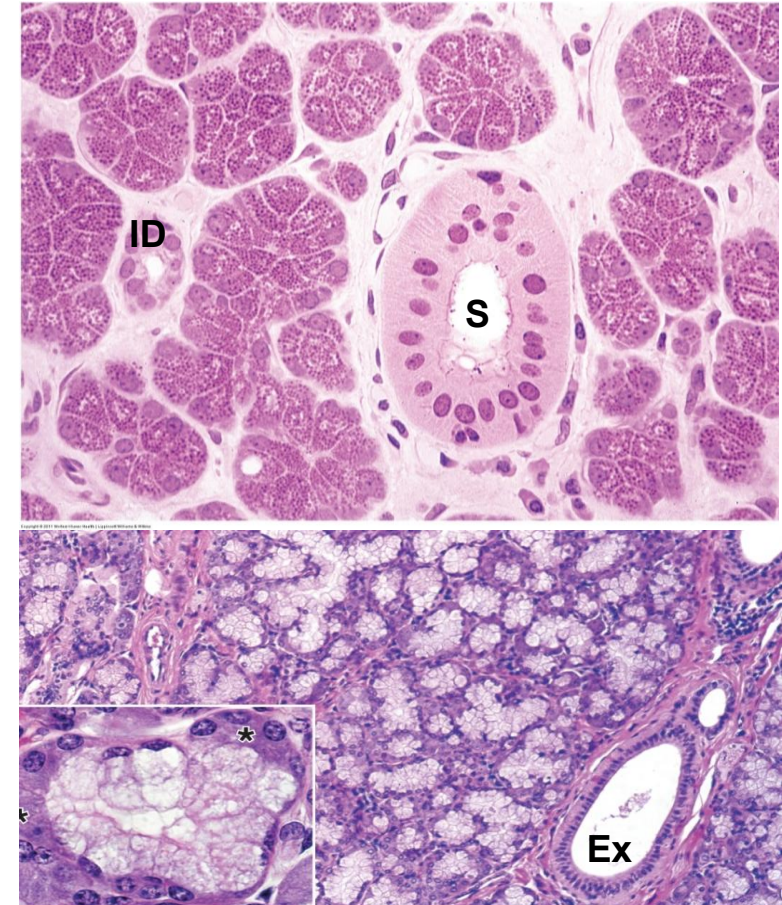


Mixed Unit

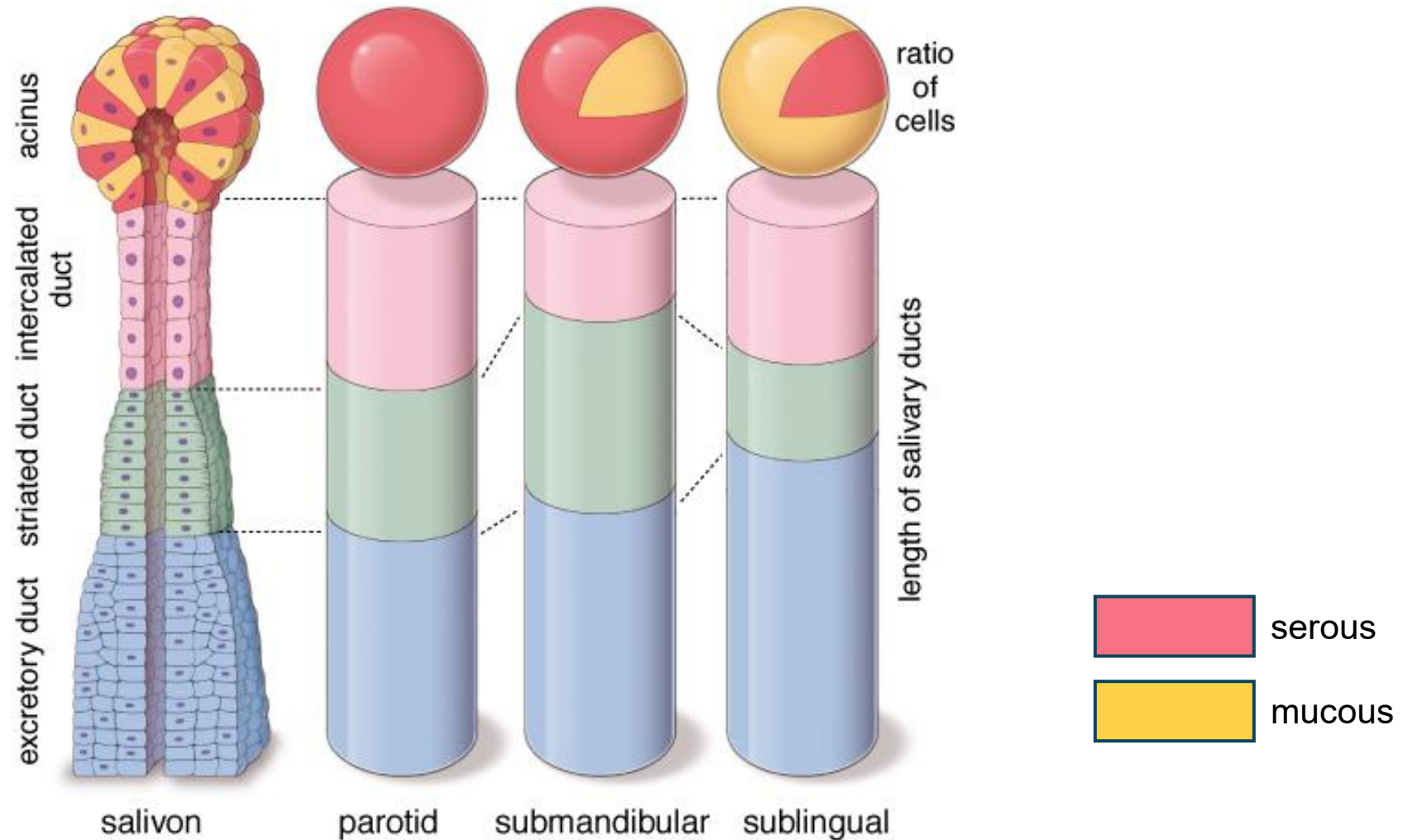


Salivary Ducts

- Three sequential segments:
 - **Intercalated ducts** → **Striated Ducts** → **Excretory ducts**
 - The proximal part of the salivary ductal system modify initial secretion and are located between the parenchyma or acini
- 1. Intercalated ducts (ID):** leads from the acinus
 - Location: Intralobular
 - Lined by a simple low cuboidal epithelium
 - Secrete HCO_3^- and absorb Cl^- from primary secretion
 - 2. Striated ducts (S)**
 - Location: Intralobular
 - Lined by simple high cuboidal to columnar epithelium
 - Basal striations → in-foldings of the basal plasma membrane, which compartmentalize mitochondria
 - Apical nuclei
 - Reabsorb Na^+ from primary secretions
 - Secretion of K^+ and adds HCO_3^- into secretions
 - 3. Excretory ducts (Ex):** empties into the oral cavity
 - Location: interlobular and interlobar connective tissue
 - Epithelium: simple cuboidal / columnar which gradually changes to pseudostratified columnar, stratified cuboidal or columnar and even stratified squamous epithelium near the oral cavity



Salivon: Acinus & Ducts



Parotid Gland

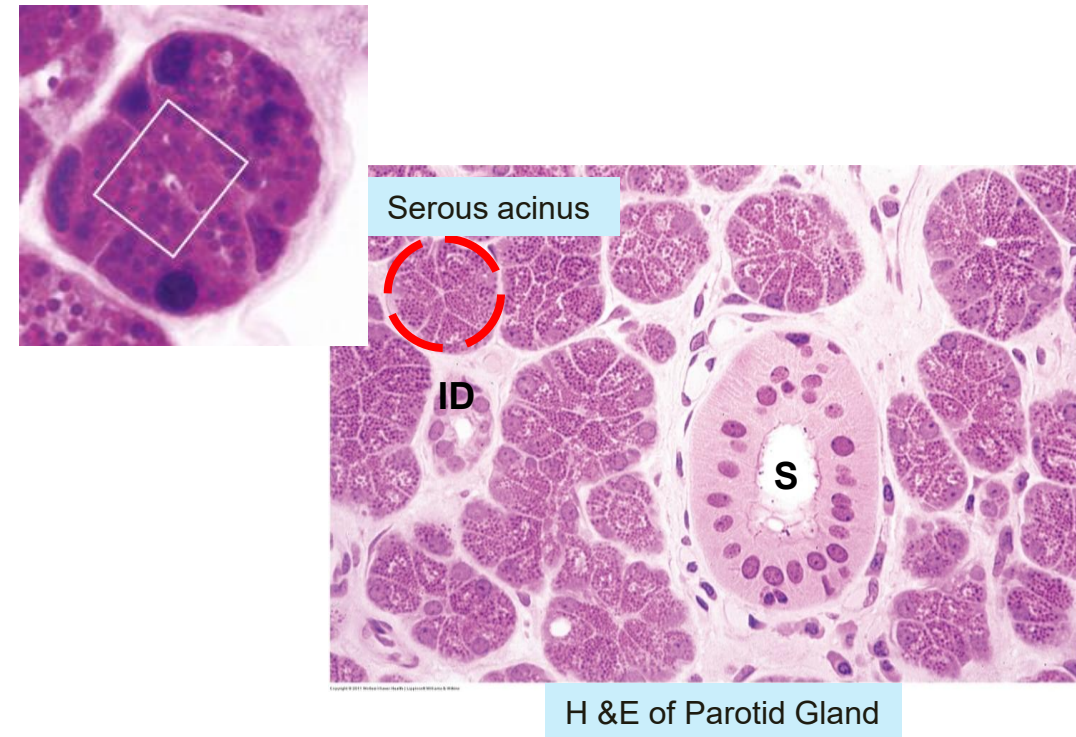
- Located anterior and inferior to the ear
- **Classified as a pure serous, compound, alveolar gland**
- Has a capsule formed by the superficial cervical fascia → dense connective tissue which forms of trabeculae (septa) that subdivide the gland into lobes and lobules
- Trabeculae convey blood and lymph vessels, excretory ducts, and nerves through the substance of the gland

Secretory portion

- Serous acini- serous cells produce amylase
- Apical secretory granules
- Granules also contain peroxidase, lysozyme, cystatins
- Myoepithelial cells around acini

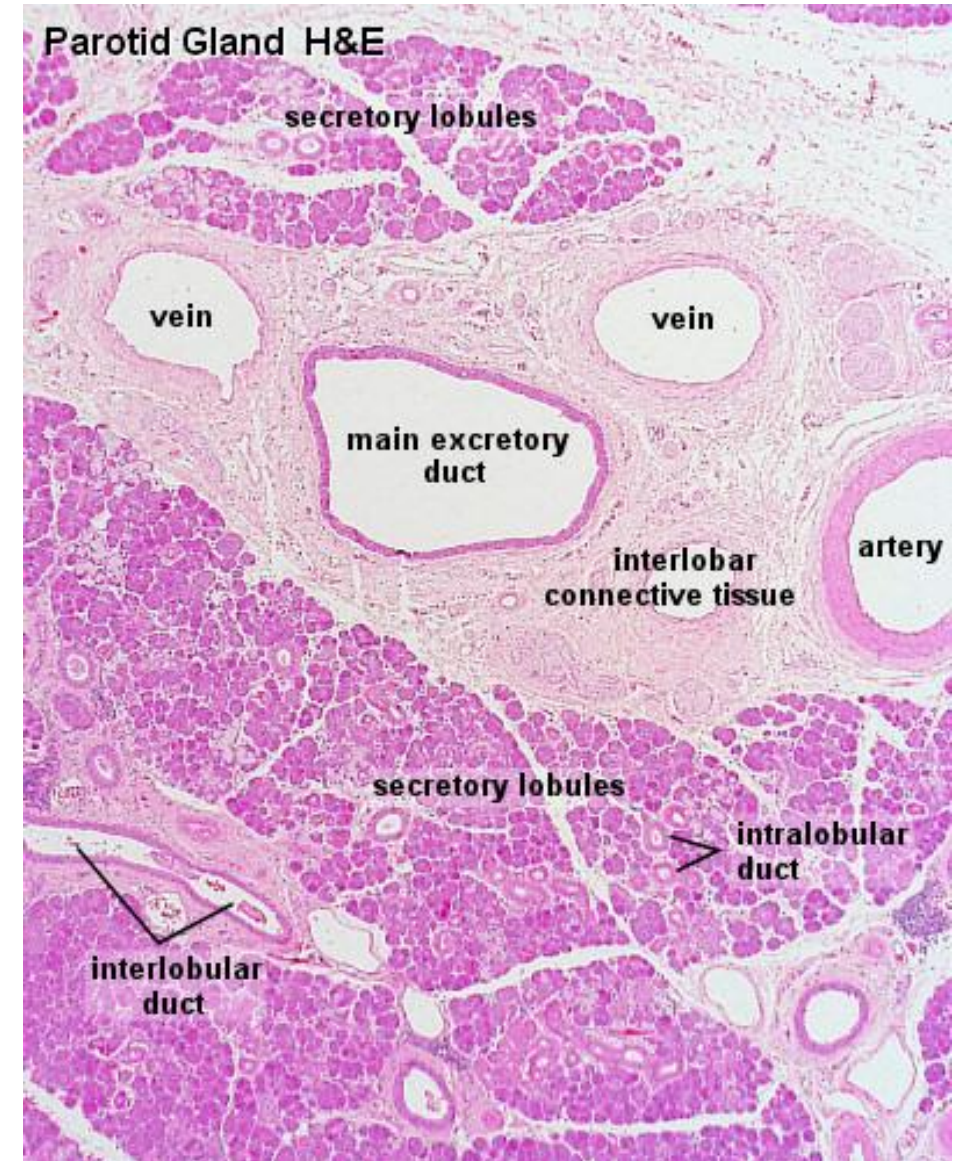
Ducts

- Intercalated (ID) and striated ducts (SD)
- The parotid gland has the longest intercalated ducts



Parotid Gland

- ***Stensen's Duct***
 - Largest excretory duct (main parotid duct)
 - **Opens into the oral vestibule at the parotid papilla (next to the upper second molar tooth)**



Submandibular Gland

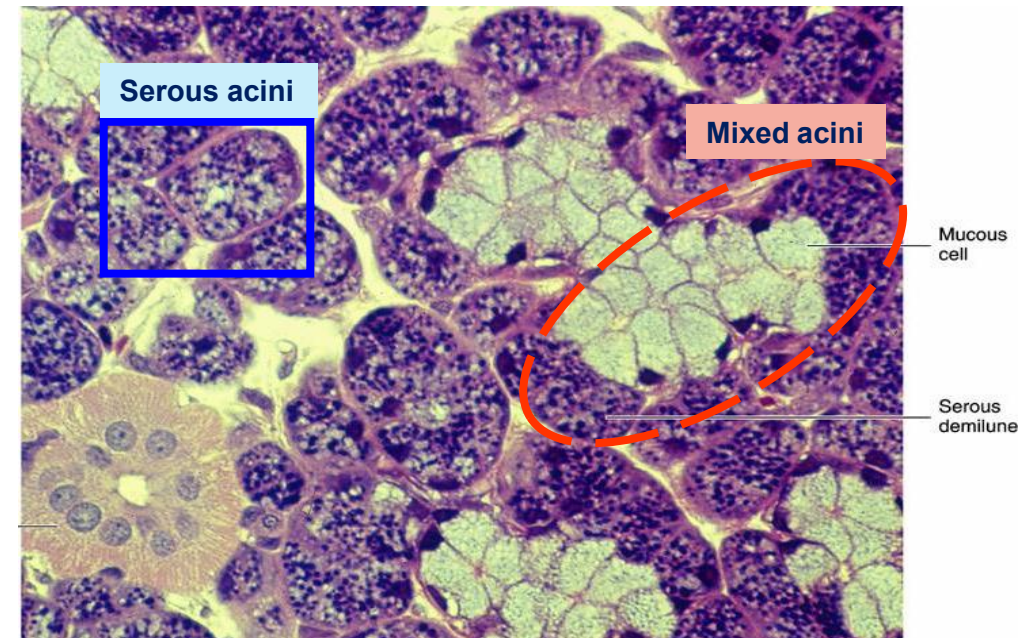
- Located on either side of the floor of the mouth close to the mandible
- **Classified as a mixed, compound, tubuloalveolar gland**

Secretory portion

- Composed mostly of pure serous acini
- However, groups of mixed acini are also present among the pure serous units
- Note the presence of dense serous cells forming demilunes and pale-staining mucous cells grouped along the tubular portion of this tubuloacinar gland
- Myoepithelial cells around acini

Ducts

- Intercalated ducts are short, but present
- Striated ducts are long and clearly evident
- **Excretory submandibular ducts open on either side of the base of the lingual frenulum**



Sublingual Gland

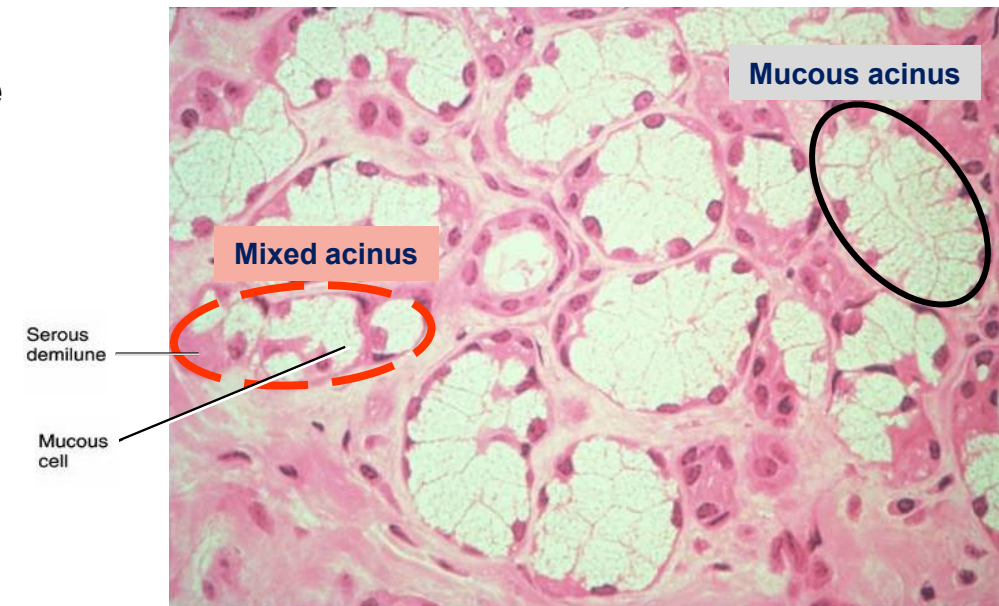
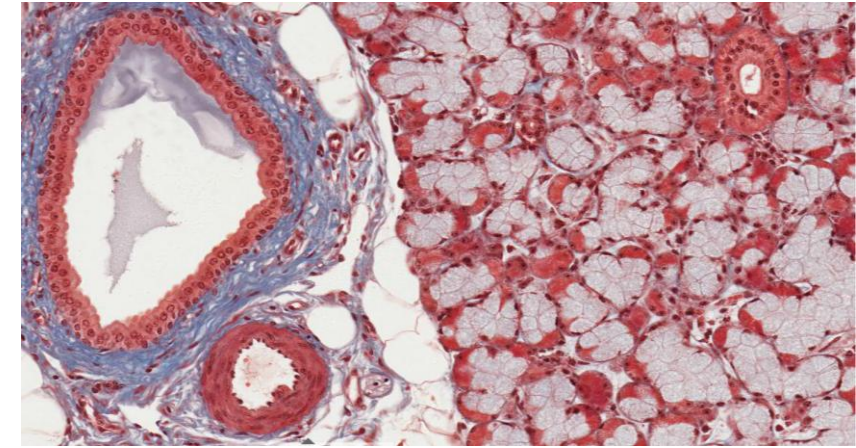
- Located on the floor of the mouth anterior and lateral to the submandibular gland
- **Classified as a mixed, compound, tubuloalveolar gland**

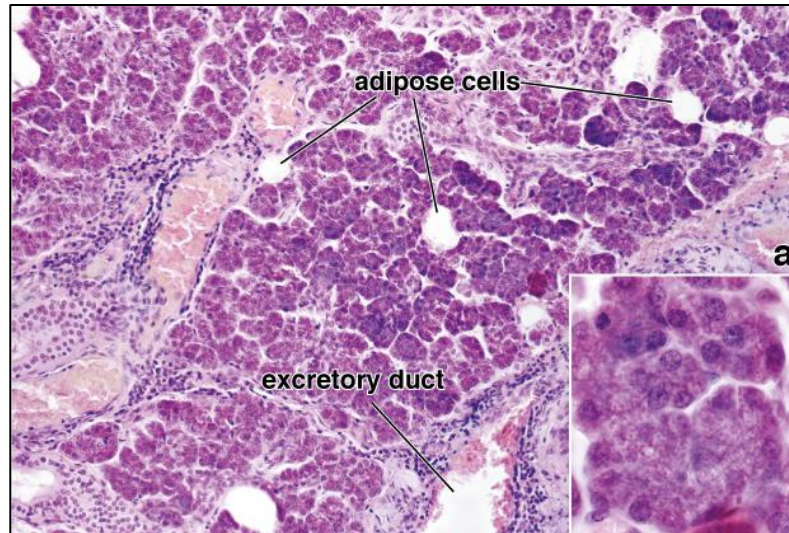
Secretory portion

- Mixed acini
- Pure mucous acinus
- Myoepithelial cells around acini

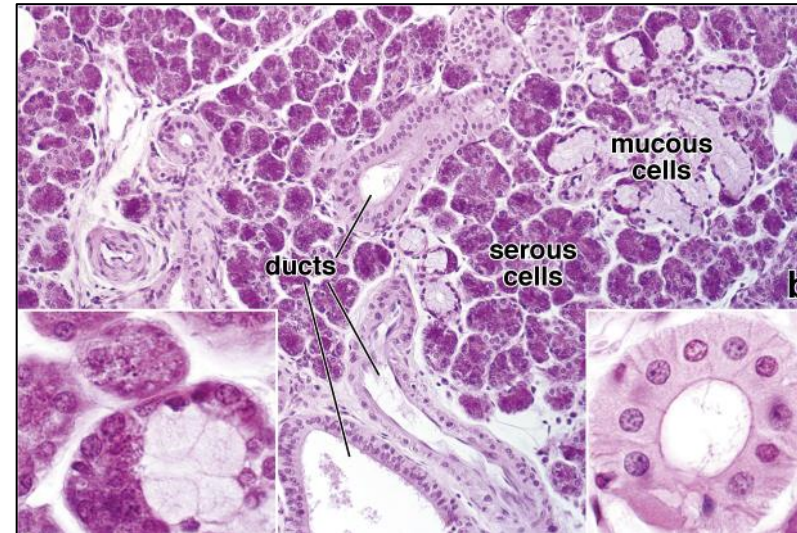
Ducts

- Short intercalated and striated ducts but are few in number
- **Multiple excretory sublingual ducts empty directly into the floor of the oral cavity and into the submandibular duct**

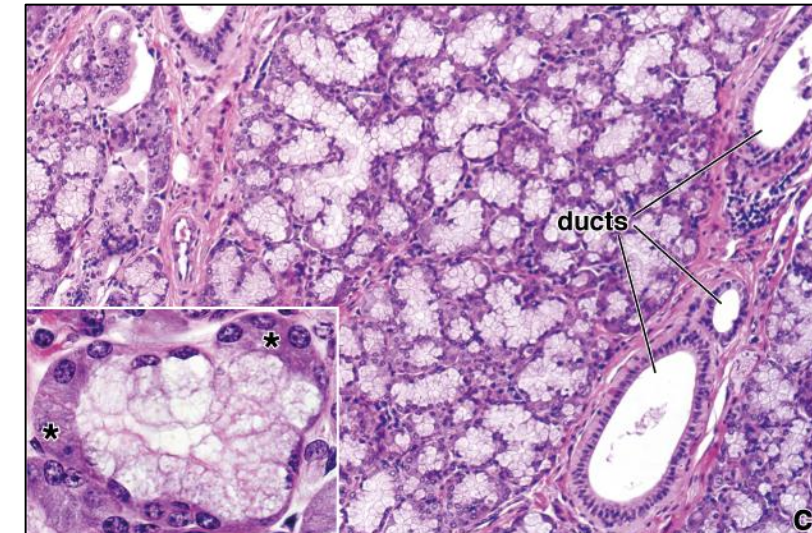




Parotid



Submandibular



Sublingual